

## Elektromagnetno polje: zbirka enačb za uporabo pri pisnih izpitih

- Potencial električnega polja in Poissonova enačba zanj

$$\mathbf{E}(\mathbf{r}) = -\nabla U(\mathbf{r}) \quad \nabla^2 U(\mathbf{r}) = -\frac{\rho(\mathbf{r})}{\varepsilon_0} \quad (1)$$

- Potencial in električno polje porazdelitve nabojev

$$U(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \int_{(V)} \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r}' \quad (2)$$

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \int_{(V)} \frac{\rho(\mathbf{r}')(\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} d^3\mathbf{r}' \quad (3)$$

- Električni dipol: potencial, električno polje in dipolni moment

$$U(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \frac{\mathbf{p}_e \cdot \mathbf{r}}{r^3} \quad (4)$$

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \frac{3\mathbf{r}(\mathbf{p}_e \cdot \mathbf{r}) - \mathbf{p}_e r^2}{r^5}. \quad (5)$$

$$\mathbf{p}_e = \int_{(V)} \mathbf{r}' \rho(\mathbf{r}') d^3\mathbf{r}' \quad (6)$$

- Električna sila na porazdelitev nabojev v zunanjem električnem polju

$$\mathbf{F}_e = \int_V \rho(\mathbf{r}') \mathbf{E}(\mathbf{r}') d^3\mathbf{r}' \quad (7)$$

- Električna sila preko napetostnega tenzorja električnega polja

$$\mathbf{F}_e = \varepsilon_0 \oint_{(\partial V)} \left[ \mathbf{E}(\mathbf{E} \cdot \mathbf{n}) - \frac{1}{2} E^2 \mathbf{n} \right] dS \quad (8)$$

- Vektorski potencial magnetnega polja in Kirchhoffova enačba zanj

$$\mathbf{B}(\mathbf{r}) = \nabla \times \mathbf{A}(\mathbf{r}) \quad \nabla^2 \mathbf{A}(\mathbf{r}) = -\mu_0 \mathbf{j}(\mathbf{r}) \quad (9)$$

- Biot-Savartov zakon za vektorski potencial in magnetno polje porazdelitve tokov

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_{(V)} \frac{\mathbf{j}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r}' \quad (10)$$

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_{(V)} \frac{\mathbf{j}(\mathbf{r}') \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} d^3\mathbf{r}' \quad (11)$$

- Biot-Savartov zakon za vektorski potencial in magnetno polje tokovnega vodnika

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \int \frac{d\mathbf{l}'}{|\mathbf{r} - \mathbf{r}'|} \quad (12)$$

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \int \frac{d\mathbf{l}' \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} \quad (13)$$

- Magnetni dipol: potencial, magnetno polje in dipolni moment

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{p}_m \times \mathbf{r}}{r^3} \quad (14)$$

$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{3\mathbf{r}(\mathbf{p}_m \cdot \mathbf{r}) - \mathbf{p}_m r^2}{r^5} \quad (15)$$

$$\mathbf{p}_m = \frac{1}{2} \int_{(V)} \mathbf{r}' \times \mathbf{j}(\mathbf{r}') d^3\mathbf{r}' \quad (16)$$

- Magnetna sila na tokovno porazdelitev v zunanjem magnetnem polju

$$\mathbf{F}_m = \int_V \mathbf{j}(\mathbf{r}') \times \mathbf{B}(\mathbf{r}') d^3\mathbf{r}' \quad (17)$$

- Magnetna sila preko napetostnega tenzorja magnetnega polja

$$\mathbf{F}_m = \frac{1}{\mu_0} \oint_{(\partial V)} \left[ \mathbf{B}(\mathbf{B} \cdot \mathbf{n}) - \frac{1}{2} B^2 \mathbf{n} \right] dS \quad (18)$$

- Kontinuitetna enačba

$$\nabla \cdot \mathbf{j} = -\frac{\partial \rho}{\partial t} \quad (19)$$

- Ohmov zakon

$$\mathbf{j} = \sigma \mathbf{E} \quad (20)$$